MOBSF

ANDROID STATIC ANALYSIS REPORT

Android Security Testing (1.3)

File Name:	hpAndro_1.3.apk
Package Name:	hpandro.android.security
Scan Date:	Oct. 31, 2026, 7:21 a.m.
App Security Score:	29/100 (CRITICAL RISK)
Grade:	F
Trackers Detection:	3/432

FINDINGS SEVERITY

HIGH	MEDIUM	INFO	SECURE	HOTSPOT
10	25	5	5	2

FILE INFORMATION

File Name: hpAndro_1.3.apk
Size: 19.3MB
MD5: c5d6f5ee36329bc922687f9966e7edf3
SHA1: 9a713b8f379068731f754896c6df44191466779a
SHA256: b014d694c58049e2e4bb1290f0cc37c4ffc5ad7e3efa554e3bf146ed1498d97a

APP INFORMATION

App Name: Android Security Testing
Package Name: hpandro.android.security
Main Activity: hpandro.android.security.ui.activity.SplashActivity
Target SDK: 30
Min SDK: 16
Max SDK:
Android Version Name: 1.3
Android Version Code: 26

APP COMPONENTS

Activities: 106
Services: 19
Receivers: 14
Providers: 4
Exported Activities: 2
Exported Services: 1
Exported Receivers: 4
Exported Providers: 0

CERTIFICATE INFORMATION

Binary is signed
v1 signature: True
v2 signature: True
v3 signature: False
v4 signature: False
v3.1 signature: False
v3.2 signature: False
X.509 Subject: CN=AppSec2022

Signature Algorithm: rsassa_pkcs1v15
Valid From: 2022-04-26 09:45:44+00:00
Valid To: 2047-04-20 09:45:44+00:00
Issuer: CN=AppSec2022
Serial Number: 0x75337007
Hash Algorithm: sha256
md5: 8eed1fb7c3eefdb9d842b15a5e922a85
sha1: e9253090f62c9b4630be955a177cd54482b4a7ae
sha256: 694d5b16da4acef57095e6ecc74afa25d7b0276b3aa2baf218be5c14cb98c764
sha512: 575225c7538cecf9ce73d8c89a2867a1c7d0739fa0e25b3c96de5471d13d6deb04f7a5eab63081ff900492c61b74e8e435fd3d8b9830ff841521cfde25c69564
PublicKey Algorithm: rsa
Bit Size: 2048
Fingerprint: c5d30c434aa516e6f92d003df98ac3f8d47c10f992575c166a3a0830ea1e9b0c
Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.INTERNET	normal	full Internet access	Allows an application to create network sockets.
android.permission.ACCESS_NETWORK_STATE	normal	view network status	Allows an application to view the status of all networks.
android.permission.ACCESS_WIFI_STATE	normal	view Wi-Fi status	Allows an application to view the information about the status of Wi-Fi.
android.permission.READ_PRIVILEGED_PHONE_STATE	unknown	Unknown permission	Unknown permission from android reference
android.permission.WRITE_EXTERNAL_STORAGE	dangerous	read/modify/delete external storage contents	Allows an application to write to external storage.
android.permission.READ_EXTERNAL_STORAGE	dangerous	read external storage contents	Allows an application to read from external storage.
android.permission.READ_PHONE_STATE	dangerous	read phone state and identity	Allows the application to access the phone features of the device. An application with this permission can determine the phone number and serial number of this phone, whether a call is active, the number that call is connected to and so on.
android.permission.ACCESS_FINE_LOCATION	dangerous	fine (GPS) location	Access fine location sources, such as the Global Positioning System on the phone, where available. Malicious applications can use this to determine where you are and may consume additional battery power.

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.ACCESS_COARSE_LOCATION	dangerous	coarse (network-based) location	Access coarse location sources, such as the mobile network database, to determine an approximate phone location, where available. Malicious applications can use this to determine approximately where you are.
android.permission.WAKE_LOCK	normal	prevent phone from sleeping	Allows an application to prevent the phone from going to sleep.
android.permission.RECEIVE_BOOT_COMPLETED	normal	automatically start at boot	Allows an application to start itself as soon as the system has finished booting. This can make it take longer to start the phone and allow the application to slow down the overall phone by always running.
hpandro.android.security.permission.C2D_MESSAGE	unknown	Unknown permission	Unknown permission from android reference
com.google.android.c2dm.permission.RECEIVE	normal	recieve push notifications	Allows an application to receive push notifications from cloud.
android.permission.VIBRATE	normal	control vibrator	Allows the application to control the vibrator.
com.sec.android.provider.badge.permission.READ	normal	show notification count on app	Show notification count or badge on application launch icon for samsung phones.
com.sec.android.provider.badge.permission.WRITE	normal	show notification count on app	Show notification count or badge on application launch icon for samsung phones.
com.htc.launcher.permission.READ_SETTINGS	normal	show notification count on app	Show notification count or badge on application launch icon for htc phones.
com.htc.launcher.permission.UPDATE_SHORTCUT	normal	show notification count on app	Show notification count or badge on application launch icon for htc phones.
com.sonyericsson.home.permission.BROADCAST_BADGE	normal	show notification count on app	Show notification count or badge on application launch icon for sony phones.
com.sonymobile.home.permission.PROVIDER_INSERT_BADGE	normal	show notification count on app	Show notification count or badge on application launch icon for sony phones.
com.anddoes.launcher.permission.UPDATE_COUNT	normal	show notification count on app	Show notification count or badge on application launch icon for apex.
com.majeur.launcher.permission.UPDATE_BADGE	normal	show notification count on app	Show notification count or badge on application launch icon for solid.
com.huawei.android.launcher.permission.CHANGE_BADGE	normal	show notification count on app	Show notification count or badge on application launch icon for huawei phones.

PERMISSION	STATUS	INFO	DESCRIPTION
com.huawei.android.launcher.permission.READ_SETTINGS	normal	show notification count on app	Show notification count or badge on application launch icon for huawei phones.
com.huawei.android.launcher.permission.WRITE_SETTINGS	normal	show notification count on app	Show notification count or badge on application launch icon for huawei phones.
android.permission.READ_APP_BADGE	normal	show app notification	Allows an application to show app icon badges.
com.oppo.launcher.permission.READ_SETTINGS	normal	show notification count on app	Show notification count or badge on application launch icon for oppo phones.
com.oppo.launcher.permission.WRITE_SETTINGS	normal	show notification count on app	Show notification count or badge on application launch icon for oppo phones.
me.everything.badger.permission.BADGE_COUNT_READ	unknown	Unknown permission	Unknown permission from android reference
me.everything.badger.permission.BADGE_COUNT_WRITE	unknown	Unknown permission	Unknown permission from android reference
com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE	normal	permission defined by google	A custom permission defined by Google.

APKID ANALYSIS

FILE	FINDINGS	DETAILS
c5d6f5ee36329bc922687f9966e7edf3.apk	Manipulator Found	Resources Confusion
c5d6f5ee36329bc922687f9966e7edf3.apk	Packer Found	DexGuard
classes3.dex	Compiler	r8 without marker (suspicious)
classes2.dex	Anti-VM Code	Build.FINGERPRINT check Build.MODEL check Build.MANUFACTURER check Build.PRODUCT check Build.TAGS check
classes2.dex	Compiler	r8 without marker (suspicious)

FILE	FINDINGS	DETAILS
classes.dex	Anti-VM Code	Build.FINGERPRINT check Build.MODEL check Build.MANUFACTURER check Build.BRAND check Build.DEVICE check Build.PRODUCT check Build.HARDWARE check Build.BOARD check possible Build.SERIAL check Build.TAGS check Build.USER check device ID check subscriber ID check ro.product.device check ro.kernel.qemu check possible ro.secure check emulator file check possible VM check
classes.dex	Anti Debug Code	Debug.isDebuggerConnected() check
classes.dex	Compiler	r8

NETWORK SECURITY

HIGH: 2 | WARNING: 2 | INFO: 0 | SECURE: 1

NO	SCOPE	SEVERITY	DESCRIPTION
1	hpandro.raviramesh.info CTF.hpandro.raviramesh.info	high	Domain config is insecurely configured to permit clear text traffic to these domains in scope.
2	hpandro.raviramesh.info CTF.hpandro.raviramesh.info	warning	Domain config is configured to trust system certificates.
3	hpandro.raviramesh.info CTF.hpandro.raviramesh.info	high	Domain config is configured to trust user installed certificates.
4	task.hpandro.raviramesh.info	secure	Domain config is securely configured to disallow clear text traffic to these domains in scope.
5	task.hpandro.raviramesh.info	warning	Domain config is configured to trust system certificates.

CERTIFICATE ANALYSIS

HIGH: 0 | WARNING: 1 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate
Application vulnerable to Janus Vulnerability	warning	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.

MANIFEST ANALYSIS

HIGH: 2 | WARNING: 9 | INFO: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 4.1-4.1.2, [minSdk=16]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Clear text traffic is Enabled For App[android:usesCleartextTraffic=true]	high	The app intends to use cleartext network traffic, such as cleartext HTTP, FTP stacks, DownloadManager, and MediaPlayer. The default value for apps that target API level 27 or lower is "true". Apps that target API level 28 or higher default to "false". The key reason for avoiding cleartext traffic is the lack of confidentiality, authenticity, and protections against tampering; a network attacker can eavesdrop on transmitted data and also modify it without being detected.
3	App has a Network Security Configuration[android:networkSecurityConfig=@xml/network_security_config]	info	The Network Security Configuration feature lets apps customize their network security settings in a safe, declarative configuration file without modifying app code. These settings can be configured for specific domains and for a specific app.
4	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.
5	Activity (hpandro.android.security.ui.activity.task.datastorage.HiddenActivity) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.
6	Service (androidx.work.impl.background.systemjob.SystemJobService) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.BIND_JOB_SERVICE [android:exported=true]	warning	A Service is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.

NO	ISSUE	SEVERITY	DESCRIPTION
7	Broadcast Receiver (com.onesignal.GcmBroadcastReceiver) is Protected by a permission, but the protection level of the permission should be checked. Permission: com.google.android.c2dm.permission.SEND [android:exported=true]	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
8	Activity (com.onesignal.NotificationOpenedActivityHMS) is not Protected. An intent-filter exists.	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Activity is explicitly exported.
9	Broadcast Receiver (com.onesignal.BootUpReceiver) is not Protected. An intent-filter exists.	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Broadcast Receiver is explicitly exported.
10	Broadcast Receiver (com.onesignal.UpgradeReceiver) is not Protected. An intent-filter exists.	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Broadcast Receiver is explicitly exported.
11	Broadcast Receiver (com.google.firebase.iid.FirebaseInstanceIdReceiver) is Protected by a permission, but the protection level of the permission should be checked. Permission: com.google.android.c2dm.permission.SEND [android:exported=true]	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
12	High Intent Priority (999) - {1} Hit(s) [android:priority]	warning	By setting an intent priority higher than another intent, the app effectively overrides other requests.

</> CODE ANALYSIS

HIGH: 6 | WARNING: 11 | INFO: 5 | SECURE: 3 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	The App logs information. Sensitive information should never be logged.	info	CWE: CWE-532: Insertion of Sensitive Information into Log File OWASP MASVS: MSTG-STORAGE-3	com/bumptech/glide/Glide.java com/bumptech/glide/disklrucache/DiskLruCache.java com/bumptech/glide/gifdecoder/GifHeaderParser.java com/bumptech/glide/gifdecoder/StandardGifDecoder.java com/bumptech/glide/load/data/AssetPathFetcher.java com/bumptech/glide/load/data/HttpUrlFetcher.java com/bumptech/glide/load/data/LocalUriFetcher.java com/bumptech/glide/load/data/mediastore/ThumbFetcher.java com/bumptech/glide/load/data/mediastore/ThumbnailStreamOpener.java com/bumptech/glide/load/engine/DecodeJob.java com/bumptech/glide/load/engine/DecodePath.java com/bumptech/glide/load/engine/Engine.java com/bumptech/glide/load/engine/GlideException.java com/bumptech/glide/load/engine/SourceGenerator.java com/bumptech/glide/load/engine/bitmap_recycle/LruArrayPool.java com/bumptech/glide/load/engine/bitmap_recycle/LruBitmapPool.java com/bumptech/glide/load/engine/cache/DiskLruCacheWrapper.java com/bumptech/glide/load/engine/cache/MemorySizeCalculator.java com/bumptech/glide/load/engine/executor/GlideExecutor.java com/bumptech/glide/load/engine/executor/RuntimeCompat.java com/bumptech/glide/load/engine/prefill/BitmapPreFillRunner.java com/bumptech/glide/load/model/ByteBufferEncoder.java com/bumptech/glide/load/model/ByteBufferFileLoader.java com/bumptech/glide/load/model/FileLoader.java com/bumptech/glide/load/model/ResourceLoader.java com/bumptech/glide/load/model/StreamEncoder.java com/bumptech/glide/load/resource/bitmap/BitmapEncoder.java com/bumptech/glide/load/resource/bitmap/DefaultImageHeaderParser.java com/bumptech/glide/load/resource/bitmap/Downsampler.java com/bumptech/glide/load/resource/bitmap/DrawableToBitmapConverter.java com/bumptech/glide/load/resource/bitmap/HardwareConfigState.java com/bumptech/glide/load/resource/bitmap/TransformationUtils.java com/bumptech/glide/load/resource/bitmap/VideoDecoder.java com/bumptech/glide/load/resource/gif/ByteBufferGifDecoder.java com/bumptech/glide/load/resource/gif/GifDrawableEncoder.java com/bumptech/glide/load/resource/gif/StreamGifDecoder.java com/bumptech/glide/manager/DefaultConnectivityMonitor.java com/bumptech/glide/manager/DefaultConnectivityMonitorFactory.java com/bumptech/glide/manager/RequestManagerFragment.java com/bumptech/glide/manager/RequestManagerRetriever.java com/bumptech/glide/manager/RequestTracker.java com/bumptech/glide/manager/SupportRequestManagerFragment.java com/bumptech/glide/module/ManifestParser.java com/bumptech/glide/request/SingleRequest.java com/bumptech/glide/request/target/CustomViewTarget.java com/bumptech/glide/request/target/ViewTarget.java com/bumptech/glide/signature/ApplicationVersionSignature.java com/bumptech/glide/util/ContentLengthInputStream.java com/bumptech/glide/util/pool/FactoryPools.java com/onesignal/AndroidSupportV4Compat.java com/onesignal/JobIntentService.java com/onesignal/OneSignal.java com/onesignal/shortcutbadger/ShortcutBadger.java hpandro/android/security/ui/activity/task/authentication/statusManipulation/StatusManiOTPActivity.java hpandro/android/security/ui/activity/task/emulatorDetection/QEMUDetectionActivity.java hpandro/android/security/ui/activity/task/logs/LogsTaskActivity.java hpandro/android/security/ui/activity/task/publicKeyPinning/PublicKeyPinning2Activity.java hpandro/android/security/ui/activity/task/rootDetection/DangerousPropsTaskActivity.java hpandro/android/security/ui/activity/task/rootDetection/RWSystemTaskActivity.java hpandro/android/security/ui/activity/task/rootDetection/RootCloakingTaskActivity.java hpandro/android/security/ui/activity/task/rootDetection/RootManagementTaskActivity.java hpandro/android/security/ui/activity/task/rootDetection/SafetyNetTaskActivity.java hpandro/android/security/ui/activity/task/websocket/WebSocketActivity.java hpandro/android/security/ui/activity/task/websocket/WebSocketSecureActivity.java hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java hpandro/android/security/utils/fcm/OSNotificationExtenderService.java hpandro/android/security/utils/fcm/OSNotificationOpenedHandler.java hpandro/android/security/utils/operation/FlagBaseActivity.java hpandro/android/security/utils/rootDetection/RootBeerNative.java hpandro/android/security/utils/rootDetection/RootDetectionUtils.java net/sqlcipher/AbstractCursor.java net/sqlcipher/BulkCursorToCursorAdaptor.java net/sqlcipher/DatabaseUtils.java net/sqlcipher/DefaultDatabaseErrorHandler.java net/sqlcipher/database/SQLiteCompiledSql.java net/sqlcipher/database/SQLiteContentHelper.java net/sqlcipher/database/SQLiteDatabase.java net/sqlcipher/database/SQLiteDebug.java net/sqlcipher/database/SQLiteOpenHelper.java net/sqlcipher/database/SQLiteProgram.java net/sqlcipher/database/SQLiteQuery.java net/sqlcipher/database/SQLiteQueryBuilder.java net/sqlcipher/database/SqliteWrapper.java org/conscrypt/Platform.java org/conscrypt/ct/CTVerifier.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
2	Weak Encryption algorithm used	high	CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-4	hpandro/android/security/ui/activity/task/encryption/DESTaskActivity.java hpandro/android/security/ui/activity/task/encryption/RC4Activity.java hpandro/android/security/ui/activity/task/encryption/ThreeDESActivity.java
3	IP Address disclosure	warning	CWE: CWE-200: Information Exposure OWASP MASVS: MSTG-CODE-2	hpandro/android/security/ui/activity/task/emulatorDetection/IPBasedCheckingActivity.java hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java org/conscrypt/TrustManagerImpl.java
4	This App uses SSL certificate pinning to detect or prevent MITM attacks in secure communication channel.	secure	OWASP MASVS: MSTG-NETWORK-4	hpandro/android/security/utils/network/APIClient.java org/conscrypt/DefaultSSLContextImpl.java org/conscrypt/SSLParametersImpl.java tech/gusavila92/apache/http/ssl/SSLContextBuilder.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
5	Files may contain hardcoded sensitive information like usernames, passwords, keys etc.	warning	CWE: CWE-312: Cleartext Storage of Sensitive Information OWASP Top 10: M9: Reverse Engineering OWASP MASVS: MSTG-STORAGE-14	com/bumptech/glide/load/Option.java com/bumptech/glide/load/engine/DataCacheKey.java com/bumptech/glide/load/engine/EngineResource.java com/bumptech/glide/load/engine/ResourceCacheKey.java com/bumptech/glide/manager/RequestManagerRetriever.java com/onesignal/GcmBroadcastReceiver.java com/onesignal/NotificationBundleProcessor.java com/onesignal/OSInAppMessageController.java com/onesignal/OSInAppMessageLocationPrompt.java com/onesignal/OSInAppMessagePrompt.java com/onesignal/OneSignalNotificationManager.java com/onesignal/OneSignalRemoteParams.java com/onesignal/WebViewManager.java hpandro/android/security/ui/activity/task/contentprovider/SQLinjectionTaskActivity.java hpandro/android/security/utils/AppConstant.java hpandro/android/security/utils/db/DataBaseHandlerKt.java hpandro/android/security/utils/flag/model/sqli/User.java io/reactivex/internal/schedulers/SchedulerPoolFactory.java org/conscrypt/OpenSSLECKeyFactory.java org/conscrypt/OpenSSLRSAKeyFactory.java
6	This App may request root (Super User) privileges.	warning	CWE: CWE-250: Execution with Unnecessary Privileges OWASP MASVS: MSTG-RESILIENCE-1	hpandro/android/security/ui/activity/task/rootDetection/RootCloakingTaskActivity.java hpandro/android/security/ui/activity/task/rootDetection/RootManagementTaskActivity.java hpandro/android/security/utils/rootDetection/RootDetectionUtils.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
7	Insecure WebView Implementation. Execution of user controlled code in WebView is a critical Security Hole.	warning	CWE: CWE-749: Exposed Dangerous Method or Function OWASP Top 10: M1: Improper Platform Usage OWASP MASVS: MSTG-PLATFORM-7	com/onesignal/WebViewManager.java
8	Remote WebView debugging is enabled.	high	CWE: CWE-919: Weaknesses in Mobile Applications OWASP Top 10: M1: Improper Platform Usage OWASP MASVS: MSTG-RESILIENCE-2	com/onesignal/WebViewManager.java
9	This App copies data to clipboard. Sensitive data should not be copied to clipboard as other applications can access it.	info	OWASP MASVS: MSTG-STORAGE-10	hpandro/android/security/utils/AppUtils.java hpandro/android/security/utils/operation/UIBaseActivity.java
10	App can write to App Directory. Sensitive Information should be encrypted.	info	CWE: CWE-276: Incorrect Default Permissions OWASP MASVS: MSTG-STORAGE-14	hpandro/android/security/MainApp.java hpandro/android/security/utils/operation/SplashBaseActivity.java
11	This App uses SQL Cipher. SQLCipher provides 256-bit AES encryption to sqlite database files.	info	OWASP MASVS: MSTG-CRYPTO-1	hpandro/android/security/ui/activity/task/datastorage/EncryptSQLiteDBActivity.java net/sqlcipher/database/SupportHelper.java
12	App uses SQLite Database and execute raw SQL query. Untrusted user input in raw SQL queries can cause SQL Injection. Also sensitive information should be encrypted and written to the database.	warning	CWE: CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') OWASP Top 10: M7: Client Code Quality	com/onesignal/OneSignalDbHelper.java com/onesignal/outcomes/OSOutcomeTableProvider.java hpandro/android/security/ui/activity/task/hash/MD4HashActivity.java hpandro/android/security/ui/activity/task/hash/MD5HashActivity.java hpandro/android/security/ui/activity/task/hash/SHA1HashActivity.java hpandro/android/security/utils/db/DataBaseHandler.java net/sqlcipher/database/SQLiteDatabase.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
13	The App uses an insecure Random Number Generator.	warning	CWE: CWE-330: Use of Insufficiently Random Values OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-6	hpandro/android/security/ui/activity/task/websocket/ws/WebSocket.java hpandro/android/security/ui/activity/task/websocket/ws/WebSocketFactory.java hpandro/android/security/utils/AppUtils.java tech/gusavila92/apache/commons/codec/digest/B64.java tech/gusavila92/apache/commons/codec/digest/UnixCrypt.java tech/gusavila92/websocketclient/WebSocketClient.java
14	MD5 is a weak hash known to have hash collisions.	warning	CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-4	hpandro/android/security/ui/activity/task/encryption/AESActivity.java hpandro/android/security/ui/activity/task/encryption/PredictableInitializationVectorActivity.java hpandro/android/security/ui/activity/task/websocket/ws/WebSocket.java hpandro/android/security/utils/AppUtils.java hpandro/android/security/utils/enc/AllEncryption.java
15	This App may have root detection capabilities.	secure	OWASP MASVS: MSTG-RESILIENCE-1	hpandro/android/security/ui/activity/task/rootDetection/SuBinaryTaskActivity.java hpandro/android/security/ui/activity/task/rootDetection/SuExistsTaskActivity.java hpandro/android/security/utils/rootDetection/RootDetectionUtils.java
16	App creates temp file. Sensitive information should never be written into a temp file.	warning	CWE: CWE-276: Incorrect Default Permissions OWASP Top 10: M2: Insecure Data Storage OWASP MASVS: MSTG-STORAGE-2	hpandro/android/security/utils/operation/FlagBaseActivity.java
17	SHA-1 is a weak hash known to have hash collisions.	warning	CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-4	hpandro/android/security/ui/activity/task/encryption/RSAActivity.java hpandro/android/security/utils/enc/GenerateRSA.java hpandro/android/security/utils/operation/FlagBaseActivity.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
18	Insecure Implementation of SSL. Trusting all the certificates or accepting self signed certificates is a critical Security Hole. This application is vulnerable to MITM attacks	high	CWE: CWE-295: Improper Certificate Validation OWASP Top 10: M3: Insecure Communication OWASP MASVS: MSTG-NETWORK-3	org/conscrypt/Conscrypt.java
19	This App uses SafetyNet API.	secure	OWASP MASVS: MSTG-RESILIENCE-7	hpandro/android/security/ui/activity/task/rootDetection/SafetyNetTaskActivity.java
20	Insecure WebView Implementation. WebView ignores SSL Certificate errors and accept any SSL Certificate. This application is vulnerable to MITM attacks	high	CWE: CWE-295: Improper Certificate Validation OWASP Top 10: M3: Insecure Communication OWASP MASVS: MSTG-NETWORK-3	hpandro/android/security/ui/activity/task/publicKeyPinning/PublicKeyPinning1Activity.java
21	The App uses the encryption mode CBC with PKCS5/PKCS7 padding. This configuration is vulnerable to padding oracle attacks.	high	CWE: CWE-649: Reliance on Obfuscation or Encryption of Security-Relevant Inputs without Integrity Checking OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-3	hpandro/android/security/ui/activity/task/encryption/AESActivity.java hpandro/android/security/ui/activity/task/encryption/PredictableInitializationVectorActivity.java
22	App can read/write to External Storage. Any App can read data written to External Storage.	warning	CWE: CWE-276: Incorrect Default Permissions OWASP Top 10: M2: Insecure Data Storage OWASP MASVS: MSTG-STORAGE-2	hpandro/android/security/ui/activity/task/datastorage/ExternalStorageActivity.java
23	Debug configuration enabled. Production builds must not be debuggable.	high	CWE: CWE-919: Weaknesses in Mobile Applications OWASP Top 10: M1: Improper Platform Usage OWASP MASVS: MSTG-RESILIENCE-2	com/bumptech/glide/BuildConfig.java com/bumptech/glide/gifdecoder/BuildConfig.java
24	Ensure that user controlled URLs never reaches the Webview. Enabling file access from URLs in WebView can leak sensitive information from the file system.	warning	CWE: CWE-200: Information Exposure OWASP Top 10: M1: Improper Platform Usage OWASP MASVS: MSTG-PLATFORM-7	hpandro/android/security/ui/activity/task/contentprovider/FileSystemExposeTaskActivity.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
25	This app listens to Clipboard changes. Some malware also listen to Clipboard changes.	info	OWASP MASVS: MSTG-PLATFORM-4	hpandro/android/security/ui/activity/task/datastorage/ClipBoardTaskActivity.java

SHARED LIBRARY BINARY ANALYSIS

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
1	arm64-v8a/libbackdoor7.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcat_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
2	arm64-v8a/libRPATH.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	hpandro{RPATH.PtVRirfn2eeX4uDcdf38F5xTimVHNXz} high The binary has RUNPATH set. In certain cases, an attacker can abuse this feature and or modify environment variables to run arbitrary libraries for code execution and privilege escalation. The only time a library should set RUNPATH is when it is linked to private libraries in the same package. Remove the compiler option --enable-new-dtags,-rpath to remove RUNPATH.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
3	arm64-v8a/libconscrypt_jni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__memmove_chk', '__strchr_chk', '__memset_chk', '__memcpy_chk', '__vsnprintf_chk', '__read_chk', '__strlen_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
4	arm64-v8a/libbackdoor6.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcat_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
5	arm64-v8a/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
6	arm64-v8a/libnativefunccall.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcat_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
7	arm64-v8a/libhardcoded secret.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
8	armeabi-v7a/libbackdoor7.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
9	armeabi-v7a/libRPATH.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	hpandro{RPATH.PtVRirfn2eeX4uDcdf38F5xTimVHNXz} high The binary has RUNPATH set. In certain cases, an attacker can abuse this feature and or modify environment variables to run arbitrary libraries for code execution and privilege escalation. The only time a library should set RUNPATH is when it is linked to private libraries in the same package. Remove the compiler option --enable-new-dtags,-rpath to remove RUNPATH.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
10	armeabi-v7a/libconscrypt_jni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
11	armeabi-v7a/libbackdoor6.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
12	armeabi-v7a/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
13	armeabi-v7a/libnativefuncall.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
14	armeabi-v7a/libhardcodedsecret.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
15	x86_64/libbackdoor7.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcat_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
16	x86_64/libRPATH.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	hpandro{RPATH.PtVRirfn2eeX4uDcdf38F5xTimVHNXz} high The binary has RUNPATH set. In certain cases, an attacker can abuse this feature and or modify environment variables to run arbitrary libraries for code execution and privilege escalation. The only time a library should set RUNPATH is when it is linked to private libraries in the same package. Remove the compiler option --enable-new-dtags,-rpath to remove RUNPATH.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
17	x86_64/libconscrypt_jni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strlen_chk', '__vsnprintf_chk', '__strchr_chk', '__memcpy_chk', '__memset_chk', '__read_chk', '__memmove_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
18	x86_64/libbackdoor6.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcat_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
19	x86_64/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
20	x86_64/libnativefuncall.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcat_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
21	x86_64/libhardcodedsecret.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
22	x86/libbackdoor7.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
23	x86/libRPATH.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	hpandro{RPATH.PtVRirfn2eeX4uDcdf38F5xTimVHNXz} high The binary has RUNPATH set. In certain cases, an attacker can abuse this feature and or modify environment variables to run arbitrary libraries for code execution and privilege escalation. The only time a library should set RUNPATH is when it is linked to private libraries in the same package. Remove the compiler option --enable-new-dtags,-rpath to remove RUNPATH.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
24	x86/libconscrypt_jni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
25	x86/libbackdoor6.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
26	x86/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
27	x86/libnativefuncall.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
28	x86/libhardcodedsecret.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

BEHAVIOUR ANALYSIS

RULE ID	BEHAVIOUR	LABEL	FILES
00077	Read sensitive data(SMS, CALLLOG, etc)	collection sms calllog calendar	com/bumptech/glide/load/data/mediastore/ThumbFetcher.java com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00146	Get the network operator name and IMSI	telephony collection	hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java

RULE ID	BEHAVIOUR	LABEL	FILES
00078	Get the network operator name	collection telephony	com/onesignal/OSUtils.java hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java
00038	Query the phone number	collection	hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java
00171	Compare network operator with a string	network	hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java
00013	Read file and put it into a stream	file	com/bumptech/glide/disklrucache/DiskLruCache.java com/bumptech/glide/load/model/FileLoader.java hpandro/android/security/ui/activity/task/emulatorDetection/QEMUDetectionActivity.java hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java okio/Okio__JvmOkioKt.java org/conscrypt/DefaultSSLContextImpl.java org/conscrypt/FileClientSessionCache.java org/conscrypt/KeyManagerFactoryImpl.java tech/gusavila92/apache/http/entity/FileEntity.java tech/gusavila92/apache/http/ssl/SSLContextBuilder.java
00117	Get the IMSI and network operator name	telephony collection	hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java
00033	Query the IMEI number	collection	hpandro/android/security/ui/activity/task/deviceID/IMEITaskActivity.java hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java
00067	Query the IMSI number	collection	hpandro/android/security/utils/emulatorDetection/EmulatorDetector.java
00089	Connect to a URL and receive input stream from the server	command network	com/bumptech/glide/load/data/HttpUrlFetcher.java com/onesignal/OneSignalRestClient.java
00030	Connect to the remote server through the given URL	network	com/bumptech/glide/load/data/HttpUrlFetcher.java
00109	Connect to a URL and get the response code	network command	com/bumptech/glide/load/data/HttpUrlFetcher.java com/onesignal/OneSignalRestClient.java

RULE ID	BEHAVIOUR	LABEL	FILES
00091	Retrieve data from broadcast	collection	com/onesignal/BundleCompatBundle.java com/onesignal/GcmBroadcastReceiver.java com/onesignal/NotificationExtenderService.java com/onesignal/OSNotificationFormatHelper.java
00163	Create new Socket and connecting to it	socket	org/conscrypt/AbstractConscryptSocket.java org/conscrypt/KitKatPlatformOpenSSLSocketImplAdapter.java org/conscrypt/PreKitKatPlatformOpenSSLSocketImplAdapter.java tech/gusavila92/apache/http/impl/pool/BasicConnFactory.java tech/gusavila92/websocketclient/WebSocketClient.java
00125	Check if the given file path exist	file	hpandro/android/security/ui/activity/task/emulatorDetection/FileBasedCheckingActivity.java hpandro/android/security/ui/activity/task/emulatorDetection/QEMUDetectionActivity.java hpandro/android/security/ui/activity/task/rootDetection/BusyBoxBinaryTaskActivity.java hpandro/android/security/ui/activity/task/rootDetection/SuBinaryTaskActivity.java
00063	Implicit intent(view a web page, make a phone call, etc.)	control	com/karumi/dexter/listener/SettingsClickListener.java com/onesignal/OSUtils.java com/onesignal/PermissionsActivity.java com/onesignal/shortcutbadger/impl/OPPOHomeBader.java com/onesignal/shortcutbadger/impl/SonyHomeBadger.java hpandro/android/security/utils/fragment/HomeWebViewFragment.java
00036	Get resource file from res/raw directory	reflection	com/karumi/dexter/listener/SettingsClickListener.java com/onesignal/OSUtils.java com/onesignal/shortcutbadger/impl/EverythingMeHomeBadger.java com/onesignal/shortcutbadger/impl/HuaweiHomeBadger.java com/onesignal/shortcutbadger/impl/NovaHomeBadger.java com/onesignal/shortcutbadger/impl/OPPOHomeBader.java com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java com/onesignal/shortcutbadger/impl/SonyHomeBadger.java
00096	Connect to a URL and set request method	command network	com/onesignal/OneSignalRestClient.java
00051	Implicit intent(view a web page, make a phone call, etc.) via setData	control	com/onesignal/OSUtils.java com/onesignal/PermissionsActivity.java hpandro/android/security/utils/fragment/HomeWebViewFragment.java

RULE ID	BEHAVIOUR	LABEL	FILES
00162	Create InetSocketAddress object and connecting to it	socket	org/conscrypt/AbstractConscryptSocket.java tech/gusavila92/apache/http/impl/pool/BasicConnFactory.java tech/gusavila92/websocketclient/WebSocketClient.java
00012	Read data and put it into a buffer stream	file	org/conscrypt/DefaultSSLContextImpl.java
00009	Put data in cursor to JSON object	file	com/onesignal/GenerateNotification.java com/onesignal/NotificationBundleProcessor.java com/onesignal/NotificationOpenedProcessor.java com/onesignal/NotificationSummaryManager.java com/onesignal/OneSignal.java
00177	Check if permission is granted and request it	permission	com/onesignal/AndroidSupportV4Compat.java
00189	Get the content of a SMS message	sms	com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00188	Get the address of a SMS message	sms	com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00011	Query data from URI (SMS, CALLLOGS)	sms calllog collection	com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00191	Get messages in the SMS inbox	sms	com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00200	Query data from the contact list	collection contact	com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00187	Query a URI and check the result	collection sms calllog calendar	com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00201	Query data from the call log	collection calllog	com/onesignal/shortcutbadger/impl/SamsungHomeBadger.java
00083	Query the IMEI number	collection telephony	hpandro/android/security/ui/activity/task/deviceID/IMEITaskActivity.java
00075	Get location of the device	collection location	hpandro/android/security/utils/LocationTrack.java
00023	Start another application from current application	reflection control	com/onesignal/OneSignal.java
00130	Get the current WIFI information	wifi collection	hpandro/android/security/ui/activity/task/deviceID/DeviceMacTaskActivity.java

FIREBASE DATABASES ANALYSIS

TITLE	SEVERITY	DESCRIPTION
Firebase Remote Config disabled	secure	Firebase Remote Config is disabled for https://firebaseremoteconfig.googleapis.com/v1/projects/568710642376/namespaces/firebase:fetch?key=AIzaSyCV7vzz3BKY7JVtpS5h-sAAwDTBn0LePaU. This is indicated by the response: {'state': 'NO_TEMPLATE'}

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	11/25	android.permission.INTERNET, android.permission.ACCESS_NETWORK_STATE, android.permission.ACCESS_WIFI_STATE, android.permission.WRITE_EXTERNAL_STORAGE, android.permission.READ_EXTERNAL_STORAGE, android.permission.READ_PHONE_STATE, android.permission.ACCESS_FINE_LOCATION, android.permission.ACCESS_COARSE_LOCATION, android.permission.WAKE_LOCK, android.permission.RECEIVE_BOOT_COMPLETED, android.permission.VIBRATE
Other Common Permissions	2/44	com.google.android.c2dm.permission.RECEIVE, com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE

Malware Permissions:
Top permissions that are widely abused by known malware.

Other Common Permissions:
Permissions that are commonly abused by known malware.

OFAC SANCTIONED COUNTRIES

This app may communicate with the following OFAC sanctioned list of countries.

DOMAIN	COUNTRY/REGION

DOMAIN MALWARE CHECK

DOMAIN	STATUS	GEOLOCATION
stackoverflow.com	ok	**IP:** 198.252.206.1 **Country:** United States of America **Region:** New York **City:** New York City **Latitude:** 40.707741 **Longitude:** -74.003273 **View:** Google Map

DOMAIN	STATUS	GEOLOCATION
nibarius.github.io	ok	**IP:** 185.199.110.153 **Country:** United States of America **Region:** Pennsylvania **City:** California **Latitude:** 40.065632 **Longitude:** -79.891708 **View:** Google Map
datagenetics.com	ok	**IP:** 172.67.222.18 **Country:** Japan **Region:** Tokyo **City:** Tokyo **Latitude:** 35.689507 **Longitude:** 139.691696 **View:** Google Map
codeshare.frida.re	ok	**IP:** 3.33.161.45 **Country:** United States of America **Region:** Washington **City:** Seattle **Latitude:** 47.627499 **Longitude:** -122.346199 **View:** Google Map
www.unknowncheats.me	ok	**IP:** 104.26.12.251 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map

DOMAIN	STATUS	GEOLOCATION
ctf.hpandro.raviramesh.info	ok	**IP:** 65.38.98.136 **Country:** United States of America **Region:** Texas **City:** Houston **Latitude:** 29.779779 **Longitude:** -95.605133 **View:** Google Map
developer.android.com	ok	**IP:** 142.250.207.206 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
youtu.be	ok	**IP:** 192.178.211.136 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
www.zetetic.net	ok	**IP:** 13.227.249.91 **Country:** Singapore **Region:** Singapore **City:** Singapore **Latitude:** 1.289670 **Longitude:** 103.850067 **View:** Google Map

DOMAIN	STATUS	GEOLOCATION
mobile-security.gitbook.io	ok	**IP:** 104.18.40.47 **Country:** United States of America **Region:** Texas **City:** Dallas **Latitude:** 32.783058 **Longitude:** -96.806671 **View:** Google Map
stuff.mit.edu	ok	**IP:** 18.4.86.31 **Country:** United States of America **Region:** Massachusetts **City:** Cambridge **Latitude:** 42.365078 **Longitude:** -71.104523 **View:** Google Map
books.nowsecure.com	ok	No Geolocation information available.
medium.com	ok	**IP:** 162.159.153.4 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map
task.hpandro.raviramesh.info	ok	**IP:** 65.38.98.124 **Country:** United States of America **Region:** Texas **City:** Houston **Latitude:** 29.779779 **Longitude:** -95.605133 **View:** Google Map

DOMAIN	STATUS	GEOLOCATION
github.com	ok	**IP:** 20.207.73.82 **Country:** United States of America **Region:** Washington **City:** Redmond **Latitude:** 47.682899 **Longitude:** -122.120903 **View:** Google Map
onesignal.com	ok	**IP:** 104.16.160.145 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map
wiki.sei.cmu.edu	ok	**IP:** 147.72.252.206 **Country:** United States of America **Region:** Pennsylvania **City:** Pittsburgh **Latitude:** 40.444321 **Longitude:** -79.954918 **View:** Google Map
en.wikipedia.org	ok	**IP:** 103.102.166.224 **Country:** United States of America **Region:** Indiana **City:** Francisco **Latitude:** 38.333332 **Longitude:** -87.447220 **View:** Google Map

DOMAIN	STATUS	GEOLOCATION
play.google.com	ok	**IP:** 142.251.151.7 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
cmrodriguez.me	ok	**IP:** 185.199.110.153 **Country:** United States of America **Region:** Pennsylvania **City:** California **Latitude:** 40.065632 **Longitude:** -79.891708 **View:** Google Map
resources.infosecinstitute.com	ok	**IP:** 172.64.154.200 **Country:** United States of America **Region:** Texas **City:** Dallas **Latitude:** 32.783058 **Longitude:** -96.806671 **View:** Google Map
android.googlesource.com	ok	**IP:** 192.178.211.82 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map

DOMAIN	STATUS	GEOLOCATION
hpandro.raviramesh.info	ok	**IP:** 65.38.98.124 **Country:** United States of America **Region:** Texas **City:** Houston **Latitude:** 29.779779 **Longitude:** -95.605133 **View:** Google Map
api.onesignal.com	ok	**IP:** 104.16.160.145 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map
gitlab.kitware.com	ok	**IP:** 66.162.65.214 **Country:** United States of America **Region:** New York **City:** Saratoga Springs **Latitude:** 43.083130 **Longitude:** -73.784569 **View:** Google Map

EMAILS

EMAIL	FILE
hpandro1337@gmail.com ĥþåñðŕö1337@ĝḿåîḽ.çöḿ	Android String Resource
appro@openssl.org	lib/arm64-v8a/libconscrypt_jni.so

TRACKERS

TRACKER	CATEGORIES	URL
Google AdMob	Advertisement	https://reports.exodus-privacy.eu.org/trackers/312

TRACKER	CATEGORIES	URL
Google Firebase Analytics	Analytics	https://reports.exodus-privacy.eu.org/trackers/49
OneSignal		https://reports.exodus-privacy.eu.org/trackers/193

HARDCODED SECRETS

POSSIBLE SECRETS
"_3des_enc_pass" : "3hpAndro"
"_3des_enc_pass" : "ordnAph3"
"ads_Client_Side_Validation_Bypass" : "ca-app-pub-8474731879788478/8516872707"
"ads_Client_Side_Validation_Bypass" : "7072786158/8748879781374748-bup-ppa-ac"
"ads_Default_Credentials" : "ca-app-pub-9990218882037464/3683880900"
"ads_Default_Credentials" : "0090883863/4647302888120999-bup-ppa-ac"
"ads_JWT_Misconfiguration" : "ca-app-pub-9990218882037464/7214850237"
"ads_JWT_Misconfiguration" : "7320584127/4647302888120999-bup-ppa-ac"
"ads_REST_API_HTTP_Methods" : "ca-app-pub-9990218882037464/1878763072"
"ads_REST_API_HTTP_Methods" : "2703678781/4647302888120999-bup-ppa-ac"
"ads_S3_bucket_misconfiguration" : "ca-app-pub-9990218882037464/7761645146"
"ads_S3_bucket_misconfiguration" : "6415461677/4647302888120999-bup-ppa-ac"
"aes_enc_pass" : "hpAndro"
"aes_enc_pass" : "ordnAph"
"blowfish_enc_pass" : "hpAndro"
"blowfish_enc_pass" : "ordnAph"
"des_enc_pass" : "hpAndro1"
"des_enc_pass" : "1ordnAph"
"google_api_key" : "AIzaSyCV7vzz3BKY7JVtpS5h-sAAwDTBn0LePaU"
"google_crash_reporting_api_key" : "AIzaSyCV7vzz3BKY7JVtpS5h-sAAwDTBn0LePaU"
"library_android_database_sqlcipher_authorWebsite" : "https://www.zetetic.net/sqlcipher/"

POSSIBLE SECRETS
"library_android_database_sqlcipher_authorWebsite" : "/rehpiclqs/ten.citetez.www//:sptth"
"rc4_enc_pass" : "hpAndro"
"rc4_enc_pass" : "ordnAph"
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POSSIBLE SECRETS
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POSSIBLE SECRETS
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SCAN LOGS

Timestamp	Event	Error
2026-10-31 07:21:51	Generating Hashes	OK
2026-10-31 07:21:51	Extracting APK	OK
2026-10-31 07:21:51	Unzipping	OK
2026-10-31 07:21:51	Parsing APK with androguard	OK
2026-10-31 07:21:51	Extracting APK features using aapt/aapt2	OK
2026-10-31 07:21:51	Getting Hardcoded Certificates/Keystores	OK
2026-10-31 07:21:55	Parsing AndroidManifest.xml	OK
2026-10-31 07:21:55	Extracting Manifest Data	OK
2026-10-31 07:21:55	Manifest Analysis Started	OK
2026-10-31 07:21:55	Reading Network Security config from network_security_config.xml	OK
2026-10-31 07:21:55	Parsing Network Security config	OK
2026-10-31 07:21:55	Performing Static Analysis on: Android Security Testing (hpandro.android.security)	OK
2026-10-31 07:21:56	Fetching Details from Play Store: hpandro.android.security	OK
2026-10-31 07:21:57	Checking for Malware Permissions	OK
2026-10-31 07:21:57	Fetching icon path	OK
2026-10-31 07:21:57	Library Binary Analysis Started	OK
2026-10-31 07:21:57	Analyzing lib/arm64-v8a/libbackdoor7.so	OK
2026-10-31 07:21:57	Analyzing lib/arm64-v8a/libRPATH.so	OK
2026-10-31 07:21:57	Analyzing lib/arm64-v8a/libconscrypt_jni.so	OK
2026-10-31 07:21:57	Analyzing lib/arm64-v8a/libbackdoor6.so	OK
2026-10-31 07:21:57	Analyzing lib/arm64-v8a/libsqlcipher.so	OK

Timestamp	Event	Error
2026-10-31 07:21:57	Analyzing lib/arm64-v8a/libnativefuncall.so	OK
2026-10-31 07:21:57	Analyzing lib/arm64-v8a/libhardcodedsecret.so	OK
2026-10-31 07:21:57	Analyzing lib/armeabi-v7a/libbackdoor7.so	OK
2026-10-31 07:21:57	Analyzing lib/armeabi-v7a/libRPATH.so	OK
2026-10-31 07:21:57	Analyzing lib/armeabi-v7a/libconscrypt_jni.so	OK
2026-10-31 07:21:57	Analyzing lib/armeabi-v7a/libbackdoor6.so	OK
2026-10-31 07:21:57	Analyzing lib/armeabi-v7a/libsqlcipher.so	OK
2026-10-31 07:21:57	Analyzing lib/armeabi-v7a/libnativefuncall.so	OK
2026-10-31 07:21:57	Analyzing lib/armeabi-v7a/libhardcodedsecret.so	OK
2026-10-31 07:21:57	Analyzing lib/x86_64/libbackdoor7.so	OK
2026-10-31 07:21:58	Analyzing lib/x86_64/libRPATH.so	OK
2026-10-31 07:21:58	Analyzing lib/x86_64/libconscrypt_jni.so	OK
2026-10-31 07:21:58	Analyzing lib/x86_64/libbackdoor6.so	OK
2026-10-31 07:21:58	Analyzing lib/x86_64/libsqlcipher.so	OK
2026-10-31 07:21:58	Analyzing lib/x86_64/libnativefuncall.so	OK
2026-10-31 07:21:58	Analyzing lib/x86_64/libhardcodedsecret.so	OK
2026-10-31 07:21:58	Analyzing lib/x86/libbackdoor7.so	OK
2026-10-31 07:21:58	Analyzing lib/x86/libRPATH.so	OK
2026-10-31 07:21:58	Analyzing lib/x86/libconscrypt_jni.so	OK
2026-10-31 07:21:58	Analyzing lib/x86/libbackdoor6.so	OK
2026-10-31 07:21:58	Analyzing lib/x86/libsqlcipher.so	OK
2026-10-31 07:21:58	Analyzing lib/x86/libnativefuncall.so	OK
2026-10-31 07:21:58	Analyzing lib/x86/libhardcodedsecret.so	OK
2026-10-31 07:21:58	Reading Code Signing Certificate	OK
2026-10-31 07:21:58	Running APKiD 3.2.0	OK
2026-10-31 07:22:02	Detecting Trackers	OK

Timestamp	Event	Error
2026-10-31 07:22:04	Decompiling APK to Java with JADX	OK
2026-10-31 07:23:25	Converting DEX to Smali	OK
2026-10-31 07:23:25	Code Analysis Started on - java_source	OK
2026-10-31 07:23:28	Android SBOM Analysis Completed	OK
2026-10-31 07:23:36	Android SAST Completed	OK
2026-10-31 07:23:36	Android API Analysis Started	OK
2026-10-31 07:23:38	Android API Analysis Completed	OK
2026-10-31 07:23:38	Android Permission Mapping Started	OK
2026-10-31 07:23:42	Android Permission Mapping Completed	OK
2026-10-31 07:23:42	Android Behaviour Analysis Started	OK
2026-10-31 07:23:44	Android Behaviour Analysis Completed	OK
2026-10-31 07:23:44	Extracting Emails and URLs from Source Code	OK
2026-10-31 07:23:46	Email and URL Extraction Completed	OK
2026-10-31 07:23:46	Extracting String data from APK	OK
2026-10-31 07:23:46	Extracting String data from SO	OK
2026-10-31 07:23:47	Extracting String data from Code	OK
2026-10-31 07:23:47	Extracting String values and entropies from Code	OK
2026-10-31 07:23:50	Performing Malware check on extracted domains	OK
2026-10-31 07:23:57	Saving to Database	OK

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Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

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